# A068 v0.4.0 — PKLSA Intrinsic Phase Clock Material Tubes, Full-field RPO and Floquet Falsifier

SST-Workbench | Framework v1.0.8

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**Abstract**

This version *actually evolves* a fully three-dimensional unforced incompressible Euler velocity field initialized from a SHA-256-identified native KnotPlot trefoil, using the unchanged A064 Euler solver, finite-core regularization, Fourier truncation and RK4. Reconstructed 3D vorticity yields an enstrophy-weighted shape coordinate and a separate enstrophy/energy functional; neither has an imposed frequency. A new Lagrangian torus is advected through the same RK4 stages as the actual Euler field, with local vorticity-tangency and material-volume diagnostics. Complete 3D velocity fields are searched for returns under three periodic translations and the 24 proper cubic rotations. The exact A064 tangent operator is separately finite-difference verified; Floquet Ritz prediction is strictly conditional on RPO and material-tube screening. Space/time/core refinement and framework provenance are retained. This is a **numerical model** initialized by PKLSA/KnotPlot geometry; it is *not* a confirmed source-native material vortex or an independent physical PKLSA clock. Previously inspected geometry and past diagnostics remain **unblinded development evidence**; official BLIND packaging does not retroactively restore epistemic blindness.

## 1 Identification and certification boundaries

| Field | Value |
|---|---|
| Catalog | A068 |
| Version | v0.4.0 |
| Framework | SST Falsifier Framework v1.0.8 candidate |
| Profile | `multilibrary_gpu` |
| Current authority | `EULER_MODEL_SIMULATION_ONLY` |
| Physical clock | `UNRESOLVED` |

## 2 Research question and objective

Can an autonomous, unforced 3D Euler vortex initialized from native trefoil geometry generate a reproducible intrinsic phase without inserting a clock frequency? What additional independent tests are needed to claim a *material, physical PKLSA* clock? The present version is a numerical prerequisite, not a physical verification.

## 3 Competing hypotheses and decision semantics

$H_0$: a numerical transient, slow vortex deformation, rigid-body phase, inadequate sampling, nonconvergent core/mesh effects, or operator-fitting artefact mimics a clock. $H_1$: a genuine source-native material vortex has a stable relative periodic orbit, coherent intrinsic oscillatory phase and independent withheld operator prediction. An unqualified source produces `UNRESOLVED`, not physical `FAIL`; a failed numerical discretization is not an SST-theory falsification.

## 4 Assumptions and limiting cases

The 47-point `knot_3.1.txt` geometry belongs to one KnotPlot-native family. Its numerical centerline has no authenticated Euler-equilibrium relaxation, no independently verified source-native topology and no SI radius. The original A064 centerline is placed in a periodic cubic domain of length $L = 2\pi$ after arbitrary normalization; Gaussian smoothing of its line vorticity uses $a = 0.4$ nominal units and $\Gamma = 1$ model unit. The finite-core filter is a numerical modeling assumption, *not* a derivation of canonical $r_c$. Actual 3D velocity and vorticity evolve without a driving frequency. A Hilbert-transform phase of a transient need not be a stable clock. A transported surface is material by construction but is a vortex tube only if $\boldsymbol{\omega}\cdot\mathbf{n}$ is suitably small on the surface. A low full-field relative return is a candidate, not a Newton–Krylov shooting certification.

## 5 Symbols, dimensions and units

| Symbol | Meaning | SI dimensions if calibrated |
|---|---|---|
| $\mathbf{u}, \boldsymbol{\omega}$ | velocity, curl of velocity | $LT^{-1}, T^{-1}$ |
| $L, a, \Gamma$ | box length, core width, circulation | $L, L, L^2T^{-1}$ |
| $E$ | mean energy per unit mass | $L^2T^{-2}$ |
| $H$ | mean helicity | $LT^{-2}$ |
| $Z$ | mean enstrophy | $T^{-2}$ |
| $C, \lambda_i$ | covariance and its eigenvalues | $L^2$ |
| $q_{\mathrm{sh}}, q_Z$ | two dimensionless functionals | 1 |
| $\phi, \widehat{\Omega}$ | observed phase, angular slope | $1, T^{-1}$ |
| $\mathbf{X}, \mathbf{n}$ | material nodes, unit normal | $L, 1$ |
| $T_R, g$ | relative return period, discrete/continuous symmetry | $T, 1$ |
| $M_g, \mu$ | relative monodromy, multiplier | $1, 1$ |

No dimensional conversion to s, m or kg is made. In particular no SST circulation scale or relativistic time-dilation law is used to set a frequency.

## 6 Actual continuous and numerical model

The finite-dimensional Galerkin operator implements the projected Lamb form of Euler:

$$\partial_t \mathbf{u} = \mathcal{P}[\mathbf{u}\times\boldsymbol{\omega}], \quad \boldsymbol{\omega} = \nabla\times\mathbf{u}, \quad \nabla\cdot\mathbf{u} = 0. \tag{1}$$

The original A064 code deposits a signed line-current along each polygon edge (cloud-in-cell), applies Gaussian finite-core smoothing and a two-thirds wavenumber cutoff, then constructs the divergence-free Fourier Biot–Savart velocity

$$\widehat{\mathbf{u}}_0(\mathbf{k}) = \frac{i\mathbf{k}\times\widehat{\boldsymbol{\omega}}_a(\mathbf{k})}{|\mathbf{k}|^2} \quad (\mathbf{k}\neq 0). \tag{2}$$

RK4 evolves this field without inserted phase. Conservation screens use

$$E = \frac{1}{2V}\int_V |\mathbf{u}|^2\,dV, \quad H = \frac{1}{V}\int_V \mathbf{u}\cdot\boldsymbol{\omega}\,dV, \quad Z = \frac{1}{2V}\int_V |\boldsymbol{\omega}|^2\,dV. \tag{3}$$

For the full periodic volume, use enstrophy weights to construct the periodic-centered three-dimensional covariance tensor $C$. Ordered eigenvalues define the shape scalar

$$q_{\mathrm{sh}} = \frac{\lambda_1-\lambda_2}{\operatorname{tr} C}, \quad q_{\mathrm{sh},2} = \frac{\lambda_2-\lambda_3}{\operatorname{tr} C}. \tag{4}$$

These two eigenvalue gaps are correlated. Instead of treating them as independent physical clocks, use the distinct flow functional

$$q_Z = \frac{L^2 Z}{E}. \tag{5}$$

Both $q_{\mathrm{sh}}$ and $q_Z$ are computed from each actual evolved 3D field, not from imposed oscillators. The analytic-signal phase estimator is

$$\phi_q(t) = \operatorname{unwrap} \arg\{[q(t) - \bar{q}] + i\mathcal{H}[q(t) - \bar{q}]\}, \qquad \widehat{\Omega}_q = \operatorname{slope}_W \phi_q(t). \tag{6}$$

This estimator is a screen only; it does not establish a relative periodic orbit or intrinsic clock. Thresholds were specified in method preflight: at least six unwrapped cycles; RMS phase-fit error at most 0.20 radians; instantaneous frequency CV at most 0.12; cross-frequency relative error at most 0.03. A constant shape under rigid rotation is no clock. A few cycles of a windowed transient are not a proof of stability. Time series are uniformly sampled by construction.

### New v0.4.0 material and stability operators

The finite-core tube seeded around the native trefoil is a *candidate* material surface. At every Euler RK4 substage its nodes obey

$$\frac{d\mathbf{X}}{dt} = \mathbf{u}(\mathbf{X}, t), \quad \eta_\omega = \frac{1}{A}\int_S \frac{|\boldsymbol{\omega}\cdot\mathbf{n}|}{\max(|\boldsymbol{\omega}|, \epsilon_\omega)}\, dA, \quad \delta_V = \frac{|V(t) - V(0)|}{|V(0)|}. \tag{7}$$

The prescribed screening limits are $\delta_V < 0.05$, area-mean $\eta_\omega < 0.05$, and pointwise maximum $|\widehat{\omega}\cdot\mathbf{n}| < 0.10$, at all retained snapshots. Net vorticity flux through a closed surface is *not* an admissible vortex-tube test; it vanishes even on surfaces not made of vortex lines. Initial tube parameters are fixed at 32 axial sections, 10 azimuthal sides and radius 0.12 model units.

The full-field symmetry-reduced return statistic is

$$r_{\mathrm{RPO}}(T, g) = \frac{\|\mathbf{u}(T) - g\mathbf{u}(0)\|_{L^2(V)}}{\|\mathbf{u}(0)\|_{L^2(V)}}, \qquad g \in \mathbb{T}^3 \rtimes O^+_{\mathrm{cube}}. \tag{8}$$

The implementation registers a 0.025 relative candidate screen and searches all 24 proper cubic rotations with 3D periodic translations. This is a *full 3D-field search*, not proof of a converged nonlinear RPO. Repeated returns, neutral drift modes, shooting and mesh/time convergence remain necessary.

The independent tangent equation is evaluated using the unmodified A064 Euler linearization,

$$\dot{\delta\mathbf{u}} = \mathcal{P}[\delta\mathbf{u}\times\boldsymbol{\omega} + \mathbf{u}\times(\nabla\times\delta\mathbf{u})], \qquad M_g = g^{-1}D\Phi^{T_R}(\mathbf{u}_0), \tag{9}$$

where $M_g v = \mu v$, $\Omega_F = |\arg\mu|/T_R$ and $\epsilon_F = \|M_g v - \mu v\|/\|M_g v\|$. This is an *operator* frequency; a measured $q$-phase never enters the tangent propagator. Ritz values from a five-vector fixed-seed subspace are diagnostics only. No physical Floquet frequency can be reported without a validated RPO, residual, spectral convergence and a withheld prediction comparison.

## 7 Executable algorithm and fixed run matrix

1. Verify byte-exact historical controls and A064 solver/source hashes; refuse changed or missing sources.

2. Load 47 native 3D XYZ vertices; normalize and map to the periodic box; deposit vorticity and compute solenoidal $\mathbf{u}(0)$ from A064.

3. In `BASIC`, evolve $N = 16$, $a = 0.40$, $\Delta t = 0.04$, $T = 6$, retaining 151 genuine 3D field snapshots and advected material mesh states.

4. In `FULL/CERTIFY`, evolve $N = 16$, $a = 0.40$, $\Delta t = 0.04$, $T = 20$, retaining 251 3D fields and synchronously advected material surfaces. Independently evolve $N = 24$ at identical $(a, \Delta t, T)$, $N = 16$ at $\Delta t = 0.02$, and $N = 16$ with $a = 0.32, 0.48$ at fixed $(T, \Delta t)$.

5. At every retained field, measure material-volume drift, surface vorticity tangency, and $E, H, Z$, divergence, CFL, shape tensor, $q_{\mathrm{sh}}$ and $q_Z$. Recover phase only from measured functionals; do not input a theoretical frequency.

6. Evaluate complete Euler state returns up to the 24 cubic rotations and 3D translations, retaining top RPO candidates and relative $L^2$ residues. Validate tangent-linear Euler against finite differences and a manufactured analytic two-mode Floquet oracle. Only attempt matrix-free Ritz on admissible candidates.

7. Require numerical conservation to close G2; record spatial/time/core diagnostic differences even when downstream physical gates are blocked. Do not promote G3 unless material-tube tangency, a verified nonlinear relative periodic orbit, independent providers and independently predicted frequency all qualify.

8. Let the official v1.0.8 runtime write unique BLIND ZIPs, PDFs and gate-ledger hashes; do not bypass reveal policy.

## 8 Source provenance and provider independence

`knot_3.1.txt`: SHA-256 `98a628af4ba5d09274f0a1845c351a7a3d633a39ff833a3ae0b78e351bafdabf`. The original A064 Python Euler solver is included byte-for-byte at `experiment/euler3d.py` and as historical reference. Earlier two A068 preflight JSONs and A064-v0.3.0 results are immutable controls, not independent vortex providers. The finite-core smoothing function, Fourier truncation and normalization are inherited from A064. The expected upstream hashes are enforced in G1, and mesh/core variation does not create distinct provider families. Input XYZ geometry has *unverified* physical vortex equilibrium and no SI units. A tracer torus, RPO scan, or tangent derivative constitutes a distinct numerical operator, not a distinct physical provider.

## 9 Dependency graph and physical guardrails

| Gate | Required decision | Initial status |
|---|---|---|
| G0 | Framework freeze, signed provenance path | runtime |
| G1 | Native geometry, solver and historical control hashes | source check |
| G2 | Real Euler 3D evolution plus $E, H, \nabla \cdot \mathbf{u}$, CFL | numerical PASS/FAIL |
| G3 | Six clean cycles; *material tube and RPO independently qualified* | UNRESOLVED |
| G4 | Independent withheld Euler/Floquet prediction | blocked |
| G5 | Converged space, time and core with physical state fixed | blocked |
| G6 | Explicit native backend parity/authority | blocked |
| G7 | Multiple independent qualified PKLSA providers | blocked |
| G8 | Physical mechanism conclusion if all required gates pass | blocked |
| G9 | Nonced reveal commitments on allowed frozen gate ledger | withheld |

Gate order is an explicit DAG. A negative numerical screen does not count as a physical hypothesis falsification. A gapped clock phase without an identified finite-core material vortex can *never* close G3 in this release.

## 10 Numerical control and exploratory observations

For G2 preregistered development thresholds: relative divergence $< 10^{-9}$; relative energy drift $< 10^{-5}$; relative helicity drift $< 10^{-4}$; CFL $< 0.25$; finite velocity field, and nontrivial evolution. Comparative tests use

$$\delta_Z(A,B) = \frac{|Z_A(T) - Z_B(T)|}{\max(|Z_B(T)|, 10^{-30})}. \tag{10}$$

Exploratory, **unblinded** local FULL prechecks (not an official frozen framework run):

| Case | Grid, core, $\Delta t$ | $Z(T)$ | $\lvert\Delta E/E\rvert_{\max}$ | Phase cycles |
|---|---|---|---|---|
| Base | $16, 0.40, 0.04$ | 0.064202 | $8.34 \times 10^{-10}$ | 0.646 |
| Spatial | $24, 0.40, 0.04$ | 0.121697 | $6.93 \times 10^{-9}$ | 0.615 |
| Temporal | $16, 0.40, 0.02$ | 0.064202 | $4.22 \times 10^{-11}$ | 0.646 |
| Core low | $16, 0.32, 0.04$ | 0.094750 | $1.69 \times 10^{-9}$ | 0.643 |
| Core high | $16, 0.48, 0.04$ | 0.035642 | $2.04 \times 10^{-10}$ | 0.723 |

$\delta_Z$ under time-step halving is $2.01 \times 10^{-9}$, while spatial $\delta_Z \simeq 0.472$ and core-spread $\simeq 0.921$. Thus temporal integration control succeeds but spatial/core physical convergence demonstrably **does not** qualify. Less than one observed phase cycle fails the method screen's six-cycle minimum; it cannot be extrapolated to a period by assumption. These values are preliminary numerical model diagnostics and will be recomputed on each framework run.

## 11 Backend authority

Actual field integration is the upstream NumPy/SciPy FP64 Euler simulator. Its runs are not equivalent to native C++ FP64 or Intel SYCL parity certification. Framework v1.0.8 backend configuration may disable certain gates; disabled scientific gates do not pass. DD32/FP32 only screen, never physical confirmation. All core/mesh branches here share one upstream mathematical method and source provenance.

## 12 Blindness, frozen protocol and reveal

The historical source and this exploratory run have already been inspected, so only framework *output confidentiality and audit integrity* are blind. Future provider-independent PKLSA-clock claims must be preregistered on a fresh held-out source before opening it. Frozen v1.0.8 publishes run-specific ZIP/PDF identities and archives previous output roots; `REVEAL_IF_ALLOWED` correctly refuses when a required physical G8 is `NOT_RUN_PREREQUISITE`. Never edit a frozen policy, ledger or HMAC commitment to force a reveal.

## 13 Statistics and confidence limits

The Euler trajectory is one deterministic single-provider numerical realization. Time snapshots, mesh steps and smoothing variants are *not* independent experimental replicates. A Hilbert phase slope on less than one cycle has no established asymptotic period. No physical confidence interval or p-value is computed. Discovery windows and candidate selection are fixed by code; no frequency target, SST phase law or operator eigenvalue is supplied.

## 14 Actual runtime results and retained evidence

Each official BASIC/FULL/CERTIFY pipeline computes and stores `EULER_DYNAMICS_RESULTS.json`, `EULER_3D_FIELDS_MODEL_ONLY.npz` with $t, \mathbf{u}, \mathbf{q}$ and conservation tracks, the hashed source manifest, full gate ledger and framework result PDF. The NPZ includes actual evolved velocity on the 3D grid,

not just 2D projections or A064 detector curves. Only the *primary* trajectory is retained as full fields; full diagnostics from all refinement branches are in JSON. Runtime gate ledger will be inserted by frozen framework execution.

## 15 Interpretation and next falsification step

The existence of nonlinear Euler evolution does not demonstrate a stable toroidal material boundary, spectral convergence, global RPO or an intrinsic physical clock. Future v0.4 qualification needs a source-native independent core field, advected material boundary with demonstrated low flux, full-state recurrence modulo allowed symmetries, a separately computed time-dependent Floquet/Koopman prediction and a new source/protocol commitment. Historical A064 spectral eigenpairs at $N = 8, 12, 16$ were nonconvergent and cannot be used as the predicted clock frequency here. Likewise, no comparison to Lorentz time dilation is permitted before a physical intrinsic oscillator is certified.

## 16 Reproducibility and bibliography

Install only into a *new* A068-v0.4.0 instance with the official `tools/new_falsifier.py` from locally present v1.0.8. Run `pytest -q tests`, then `run_all.cmd FREEZE`, `run_all.cmd BASIC`, `run_all.cmd FULL`, and optionally `run_all.cmd CERTIFY`; no frozen package from v0.2.0 is overwritten. Preserve all v0.2.0 result evidence and commits.

## References


[1] P. G. Saffman, *Vortex Dynamics* (Cambridge University Press, 1992), doi:10.1017/CBO9780511624063.

[2] H. K. Moffatt, J. Fluid Mech. **35**, 117–129 (1969), doi:10.1017/S0022112069000991.

[3] D. Gabor, J. Inst. Elect. Eng. Part III **93**, 429–457 (1946), doi:10.1049/ji-3-2.1946.0074.